

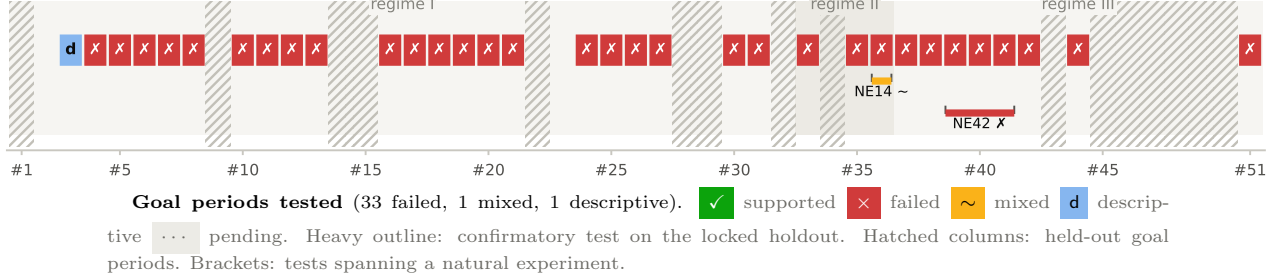
H51: One dial: an effective coupling collapses the phase diagram

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 42% Faithfulness 1.5/5 Usefulness 2/5

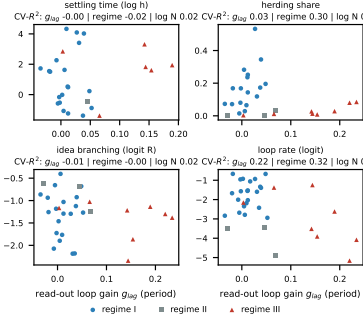
Bottom line. No single effective coupling collapses the village's phase diagram: the read-out gain is too small to matter ($\chi \leq 1.31$), and scaffold regime and swarm size carry the predictable variation.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH178 asked whether one effective coupling K puts every goal period on one curve. H67's read-out loop gain g_{lag} is that K : readers per message times an address-gated, size-diluted jump per read. H51 places 33 non-holdout periods by coupling (g_{lag}), field ($c_{\times}$, kickoff specificity) and N . It then scores four unfitted observables: settling time, herding share, idea branching and loop rate. g_{lag} collapses none of them. Its cross-validated R^2 is ≤ 0.22 , and regime labels or $\log N$ do better. The within-regime permutation gives $p = 0.07$; synthetic power is 0.89 at the effect that matters. A fitted single index is $\approx \log N$ and also collapses 0/4. In three natural experiments the observables follow N , not the dial. **The HH claim fails.** The holdout is not run.



Observables. Each point is one goal period. The x-axis is the read-out loop gain g_{lag} ; colour and marker give the scaffold regime. Regime-I periods sit at $g_{lag} \approx 0$ and spread over the whole range of every observable. Panel titles give the leave-one-period-out R^2 of g_{lag} , regime labels and $\log N$. None of the four panels lies on one curve.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Mean-field Potts/Ising (models 10, 02) near a subcritical fixed point. An agent's state (project, idea, talk) relaxes toward the local field $h_{ext} + K m$, where m is the population order parameter. K is the read-out coupling g_{lag} , h the in-window shared field (H86 $c_{\times}$, trimmed) and N the active population. One-dial collapse means that every observable Y_j is a function of one combination $u(h, K, N)$.

For an operator or a monitor: do not use the coupling dial to forecast swarm-level outcomes. Every configuration seen is subcritical ($g_{lag} \leq 0.23$), so coupling amplifies a push by at most $\times 1.3$. Observables vary $\times 3$ to $\times 127$ across periods. Two coordinates carry what is predictable: the scaffold regime (loop rate falls from 17% to 3–5%) and the swarm size. Herding falls with N ($\rho = -0.75$), and idea branching rises with N inside #51 ($\rho = 0.83$). For a phase diagram, plot N against g_{lag} with the regime as marker, and keep $c_{\times}$ as an impostor gauge, not an axis.

$$m = h_{ext} \chi, \quad \chi = \frac{1}{1 - K}, \quad \tau_{relax} = \frac{\tau_0}{1 - K}$$

$$Y_j = F_j(u), \quad u = \mathbf{w} \cdot \mathbf{z}, \quad \mathbf{z} = (g_{lag}, \text{asinh}(c_{\times}/0.01), S_{text}, \log N)$$

$$R_{cv}^2 = 1 - \sum_p (Y_p - \hat{Y}_p^{(-p)})^2 / \sum_p (Y_p - \bar{Y}^{(-p)})^2$$

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RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
S1–S3 synthetic (real axes)	size 3–4%; permutation power 0.89 at $\rho_K^2 = 0.3$; index $ \cos \geq 0.8$ in 98%	pass
P1 g_{lag} collapses $\leq 1/4$	0/4; within-regime $p = 0.073$	supported
P2 single index $\leq 2/4$	0/4; index $\approx \log N$ (weight 0.86–0.95)	supported
P3 $g_{lag} \approx$ regime labels	worse than regime on herding (–0.27), loops (–0.10)	failed
P4 $\rho(\hat{R}, g_{lag}) > 0$ in III	–0.40 [–0.86, 0.42], $n = 8$	failed
P5 no critical speed-up	$\rho(\tau, g_{lag}) = 0.05$ [–0.34, 0.42]	supported
P6 field beats g_{lag} on $\geq 1/4$; field $R_{cv}^2 \leq 0.06$		failed
P7 equal-time dial no 4/4 better		supported
Natives NE14, NE42, G51	K step barely moves loops; #40 follows N (4/4 signs); $\hat{R}$ rises with N at flat g_{lag}	mixed / failed / failed

GOAL PERIODS IN DETAIL

G03 d descriptive

G04 × failed

G05 × failed

G06 × failed

G07 × failed

G08 × failed

G10 × failed

G11 × failed

G12 × failed

G13 × failed

+25 more periods: see the card.

FAITHFULNESS SCORECARD

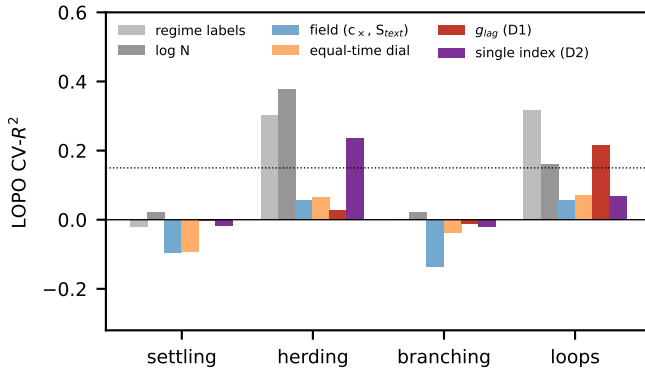
A mapping 1 F identifiability 2

B assumptions 1 G ground truth 0

C adequacy 1 H comparative 1

D unfitted predictions 1 I transfer 0

E interventional 0

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

Second figure. Leave-one-period-out R^2 per observable and model (27–33 periods). The read-out gain (red) and the single index (purple) never clear the collapse threshold (dotted, 0.15) while beating both rivals. $\log N$ wins for herding, regime labels for loops. Nothing predicts settling time.

CAVEATS

The unit is a goal period: 27–33 points per observable and only 8 in regime III. The negative rests on the permutation test and on cross-validated R^2 ; the 3/4 count rule is underpowered (18%). The observables are not impostor-clean at period level. Herding has no in-flight placebo, loop rate depends on model family, and roster composition differs between periods. In regime I, g_{lag} is measured on the wrong clock for talk. #51's size sweep is confounded with calendar time. A post hoc regime-III pattern (herding rises with g_{lag} , $\rho = 0.83$, $n = 8$) is 2 of 48 tests. Predictions were written knowing the input cards' headlines.

ROUND 2

Test the regime-III herding-coupling lead on sub-units (≥ 20 points). Fit within-regime size laws $Y \propto N^\beta$ for herding and branching. Rebuild herding and loops with convergence and family controls before any new collapse test.

- **Essence.** a low-dimensional phase diagram an operator can read. Round 1 says two coordinates (scaffold regime, N) carry what is predictable; coupling is a small correction.
- **H51-R1** Regime-III collapse with sub-units.
- **H51-R2** Size law per observable.
- **H51-R3** Impostor-clean observables.

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